

Supplementary Semester Examination – Summer 2023

Course: B. Tech. Branch: Civil Engineering Semester: VII

Subject Code & Name: BTCVC703 Water Resource Engineering

Max Marks: 60 Date: 18/07/2023 Duration: 3 Hrs.

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve the following questions.		12
A) Define base period, delta and duty. What are the factors affecting duty?	CO1	06
B) Explain classes of soil water with neat labelled sketches	CO2	06
Q.2 Solve Any Two of the following.		12
A) Define Arch dam. Describe various types of arch dam with suitable diagram.	CO2	06
B) Explain the various levels in reservoir with suitable diagram.	CO2	06
C) Write down the factors affecting selection of dam for available site conditions.	CO2	06
Q. 3 Solve Any Two of the following.		12
A) What do you mean by spillways? Why are spillways provided in dams? Enlist the types of spillways.	CO2	06
B) Explain Bligh's creep theory and state its limitations.	CO2	06
C) Why should lining be provided in canals? What are the merits and demerits of canal lining?	CO2	06
Q.4 Solve Any Two of the following.		12
A) Explain the types of tube wells with neat sketches.	CO1	06
B) Define the following terms:		
1. Specific yield 4. Yield of well	CO1	06
2. Unit Hydrograph 5. Catchment area		
3. Aquifuges 6. Infiltration		
C) What are the different types of precipitation? Describe with neat sketches.	CO 2	06
Q. 5 Solve the following questions.		12
A) How to reclaim the water-logged alkaline sands.	CO 1	06
B) What are the preventive and curative measures of water logging?	CO1	06

*** End ***